

Introduction to Basic and Advanced Statistical Modelling

Using R

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RStudio

Installation

- install R: <https://cran.r-project.org/>
- install RStudio Desktop: <https://posit.co/download/rstudio-desktop/>
- let us set the global options properly together

RStudio panes

The screenshot displays the RStudio IDE interface with the following panes and content:

- Editor Pane (Top Left):** Shows a script named "Untitled1" with a single line of code: `1`. The status bar indicates "1:1 (Top Level)" and "R Script".
- Environment Pane (Top Right):** Displays the "Environment" tab, showing "R" and "Global Environment". The message "Environment is empty" is centered.
- Files Pane (Bottom Right):** Shows the "Files" tab with a file explorer view of the home directory. The table below lists the files and their details:

Name	Size	Modified
.bash_history	27.1 KB	Jul 31, 2026, 10:49 AM
.bash_logout	18 B	Jul 23, 2025, 2:00 AM
.bash_profile	205 B	Mar 31, 2026, 9:11 PM
.bashrc	590 B	Apr 26, 2026, 12:56 PM
.cache		
.config		
.dropbox		
.dropbox-dist		
.duckdb		
.gitconfig	130 B	Apr 23, 2026, 10:56 PM
.gitkraken		
.gk		
.gnupg		
.gtkrc-2.0	340 B	Jul 31, 2026, 10:20 AM
.java		

- Console Pane (Bottom Left):** Shows the R version and copyright information, followed by the R license text and a prompt for help.

Practice 1

Let's setup our working environment:

1. create a new project
2. create a new R script
3. save the created R script into the project folder directory

NB:

- a project is defined by a simple (text) file. Its main benefit is to open RStudio with the correct working directory set (that is, the one where the project file is)
- an R script is a (text) file where we can write R code

Practice 2

Let's install the following R packages:

1. `{tidyverse}`
2. `{sf}`

NB:

- I will try to use `{}` consistently to indicate that something is a package
- installing a package is only required once, unless you want to update the package or unless you installed a new version of R
- I recommend using RStudio for installing packages to avoid running the installation code every time you restart your session

Reading data

Data of the Swiss common breeding bird monitoring program

Nicolas Strebel | Samuel Wechsler | Roman Bühler | Guido Häfliger |
Verena Keller | Marc Kéry  | Christian Rogenmoser | Martin Spiess |
Katarina Varga | Bernard Volet | Niklaus Zbinden | Hans Schmid

Since 1999, 267 one-km squares laid out in a mostly systematic grid across all of Switzerland have been surveyed annually by skilled, mostly volunteer ornithologists. Bird populations are recorded using a simplified territory mapping protocol with two visits per square above the timberline and three elsewhere. Surveys are conducted during the breeding season (mid-April to early July) along a square-specific transect route that does not change over the years.

Practice 3

1. download the datasets from the following location:
<https://zenodo.org/records/18480886>
2. extract the datasets into your R project within a folder called data_MHB

Practice 3

1. download the datasets from the following location:
<https://zenodo.org/records/18480886>
2. extract the datasets into your R project within a folder called data_MHB

Pro tips:

- you can use `getwd()` to check where your project folder is located
- this could be done using R code, but it is not necessary, nor useful here

Activating the {tidyverse}

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
v dplyr      1.2.1      v readr      2.2.0  
v forcats   1.0.1      v stringr   1.6.0  
v ggplot2    4.0.3      v tibble    3.3.1  
v lubridate  1.9.5      v tidyr     1.3.2  
v purrr      1.2.2  
-- Conflicts ----- tidyverse_conflicts() --  
x dplyr::filter() masks stats::filter()  
x dplyr::lag()     masks stats::lag()  
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to be
```

The {tidyverse} environment

The {tidyverse} is a meta package encapsulating an ecosystem of R packages (see <https://www.tidyverse.org> for more).

There are currently 9 core packages automatically attached by the {tidyverse}:



- philosophy: making R more accessible
- backward compatibility is not strictly enforced

Let's import the first dataset

```
species_count <- read_csv("data_MHB/header-species-count.csv")
```

```
Rows: 7113 Columns: 10
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
dbl (10): samplearea_id, x-koord, y-koord, lon, lat, year, route_length, median_elevat...
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Let's check the first dataset

```
species_count
```

```
# A tibble: 7,113 x 10
```

	samplearea_id	`x-koord`	`y-koord`	lon	lat	year	route_length	median_elevation
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	268	2725500	1182500	9.08	46.8	1999	4444	1550
2	268	2725500	1182500	9.08	46.8	2000	4444	1550
3	268	2725500	1182500	9.08	46.8	2001	4444	1550

```
# i 7,110 more rows
```

```
# i 2 more variables: n_visits <dbl>, n_recorded_species <dbl>
```

Let's check the first dataset

```
species_count
```

```
# A tibble: 7,113 x 10
```

	samplearea_id	`x-koord`	`y-koord`	lon	lat	year	route_length	median_elevation
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	268	2725500	1182500	9.08	46.8	1999	4444	1550
2	268	2725500	1182500	9.08	46.8	2000	4444	1550
3	268	2725500	1182500	9.08	46.8	2001	4444	1550

```
# i 7,110 more rows
```

```
# i 2 more variables: n_visits <dbl>, n_recorded_species <dbl>
```

Pro tips:

- check the output (a tibble) carefully to make sure you read in your data properly
- if your are not using {readr}, the output may not be a tibble and would be better checked using `str()`

Practice 4

- import the file `data-territories.csv` to obtain the second dataset

```
territories
```

```
# A tibble: 1,152,306 x 10
```

	species_id	euring_id	nameeg	namelt	samplearea_id	year	territories_visit_1
	<dbl>	<dbl>	<chr>	<chr>	<dbl>	<dbl>	<dbl>
1	50	70	Little Grebe	Tachybaptus r~	268	1999	0
2	50	70	Little Grebe	Tachybaptus r~	268	2000	0
3	50	70	Little Grebe	Tachybaptus r~	268	2001	0

```
# i 1,152,303 more rows  
# i 3 more variables: territories_visit_2 <dbl>, territories_visit_3 <dbl>,  
#   territories_total <dbl>
```

Practice 4

- import the file `data-territories.csv` to obtain the second dataset

```
territories
```

```
# A tibble: 1,152,306 x 10
```

	species_id	euring_id	nameeg	namelt	samplearea_id	year	territories_visit_1
	<dbl>	<dbl>	<chr>	<chr>	<dbl>	<dbl>	<dbl>
1	50	70	Little Grebe	Tachybaptus r~	268	1999	0
2	50	70	Little Grebe	Tachybaptus r~	268	2000	0
3	50	70	Little Grebe	Tachybaptus r~	268	2001	0

```
# i 1,152,303 more rows  
# i 3 more variables: territories_visit_2 <dbl>, territories_visit_3 <dbl>,  
#   territories_total <dbl>
```

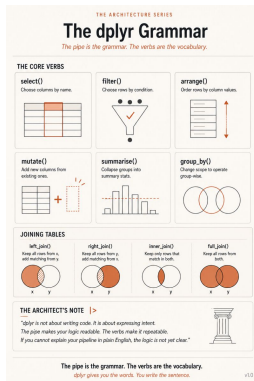
Pro tips:

- use `View()` to explore the datasets in an Excel-like fashion

Wrangling data

Data wrangling with {dplyr}

In {dplyr} much can be achieved with a few functions



source: <https://linkedin.com/in/adalbert-ngongang-phd-637995a3>

We will study some of these functions and a few others to get you going

Choose columns using `select()`

```
species_count |> select(samplearea_id, year, n_recorded_species)
```

```
# A tibble: 7,113 x 3
```

	samplearea_id	year	n_recorded_species
	<dbl>	<dbl>	<dbl>
1	268	1999	27
2	268	2000	25
3	268	2001	30

```
# i 7,110 more rows
```

Choose columns using `select()`

```
species_count |> select(samplearea_id, year, n_recorded_species)
```

```
# A tibble: 7,113 x 3
```

	samplearea_id	year	n_recorded_species
	<dbl>	<dbl>	<dbl>
1	268	1999	27
2	268	2000	25
3	268	2001	30

```
# i 7,110 more rows
```

Pro tips:

- you can use `-column_name` to drop a column called `column_name`
- you can use selection helpers (e.g. `contains()`, `starts_with()`) to select columns using name patterns
- you can use test(s) on column values to select them (`where()`)

A few words about the pipe |>

```
species_count |> select(samplearea_id, year, n_recorded_species)
```

is interpreted exactly as this:

```
select(species_count, samplearea_id, year, n_recorded_species)
```

NB:

- the pipe captures what precedes it and considers it as the first argument of the function that follows it
- it makes the code more readable, as it emphasises the sequence of operations applied to the data

Extract one column using `pull()`

You can use `select()` to choose one column, but it will remain a table (technically an object of class `tibble` or `data.frame`):

```
species_count |> select(n_recorded_species)
```

```
# A tibble: 7,113 x 1
  n_recorded_species
              <dbl>
1                  27
2                  25
3                  30
# i 7,110 more rows
```


Extract one column using `pull()`

You can use `select()` to choose one column, but it will remain a table (technically an object of class `tibble` or `data.frame`):

```
species_count |> select(n_recorded_species)
```

```
# A tibble: 7,113 x 1
```

```
  n_recorded_species
```

```
    <dbl>
```

```
1             27
```

```
2             25
```

```
3             30
```

```
# i 7,110 more rows
```

Instead `pull()` creates a vector, which can be easier to work with:

```
species_count |> pull(n_recorded_species)
```

```
[1] 27 25 30 31 29 29 38 27 29 39 30 37 44 36 37 33 35 34 42 31 42 33 37 31 32 33 42 29  
[29] 28 32 28 30 29 26 28 29 31 37 30 27 28 27 30 26 30 27 29 29 29 26 29 28 25 27 37 37  
[57] 42 46 42 41 40 41 40 42 40 45 42 41 44 40 36 34 37 36 39 43 45 41 42 44 43 38 32 34  
[85] 32 37 37 34 34 33 33 33 37 32 34 31 32 32 33 36 35 36 54 39 44 45 42 39 34 37 33 33  
[113] 33 43 41 41 37 34 42 36 39 37 34 37 40 34 38 37 36 37 39 36 31 34 30 39 41 46 42 46
```

Choose rows using `filter()`

```
species_count |> filter(year == 2025)
```

```
# A tibble: 261 x 10
```

	samplearea_id	`x-koord`	`y-koord`	lon	lat	year	route_length	median_elevation
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	268	2725500	1182500	9.08	46.8	2025	4444	1550
2	269	2497500	1118500	6.11	46.2	2025	6232	450
3	270	2725500	1214500	9.09	47.1	2025	4081	1350

```
# i 258 more rows
```

```
# i 2 more variables: n_visits <dbl>, n_recorded_species <dbl>
```

Choose rows using `filter()`

```
species_count |> filter(year == 2025)
```

```
# A tibble: 261 x 10
```

	samplearea_id	`x-koord`	`y-koord`	lon	lat	year	route_length	median_elevation
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	268	2725500	1182500	9.08	46.8	2025	4444	1550
2	269	2497500	1118500	6.11	46.2	2025	6232	450
3	270	2725500	1214500	9.09	47.1	2025	4081	1350

```
# i 258 more rows
```

```
# i 2 more variables: n_visits <dbl>, n_recorded_species <dbl>
```

Pro tips:

- use `%in%` to filter multiple rows (e.g. `year %in% c(2000, 2025)`)
- use `!=` to negate equality (e.g. `year != 2025` to keep all be 2025)

Use `filter() + if_any() + is.na()` to look for missing values

```
territories |> filter(if_any(everything(), is.na))
```

```
# A tibble: 391,247 x 10
```

	species_id	eurings_id	nameeg	namelt	samplearea_id	year	territories_visit_1
	<dbl>	<dbl>	<chr>	<chr>	<dbl>	<dbl>	<dbl>
1	50	70	Little Grebe	Tachybaptus r~	271	2001	NA
2	50	70	Little Grebe	Tachybaptus r~	271	2002	NA
3	50	70	Little Grebe	Tachybaptus r~	279	2001	NA

```
# i 391,244 more rows
```

```
# i 3 more variables: territories_visit_2 <dbl>, territories_visit_3 <dbl>,
```

```
# territories_total <dbl>
```

Use `filter() + if_any() + is.na()` to look for missing values

```
territories |> filter(if_any(everything(), is.na))
```

```
# A tibble: 391,247 x 10
```

	species_id	eurung_id	nameeg	namelt	samplearea_id	year	territories_visit_1
	<dbl>	<dbl>	<chr>	<chr>	<dbl>	<dbl>	<dbl>
1	50	70	Little Grebe	Tachybaptus r~	271	2001	NA
2	50	70	Little Grebe	Tachybaptus r~	271	2002	NA
3	50	70	Little Grebe	Tachybaptus r~	279	2001	NA

```
# i 391,244 more rows
```

```
# i 3 more variables: territories_visit_2 <dbl>, territories_visit_3 <dbl>,
```

```
# territories_total <dbl>
```

To drop NAs, you can use the {tidyr} function `drop_na()`

```
territories |> drop_na() -> territories_noNA # by default discards NA from any column
pct_lost <- 100 * (nrow(territories) - nrow(territories_noNA)) / nrow(territories)
pct_lost # percentage of rows lost b
```

```
[1] 33.95339
```

Choose rows by position using `slice()`

```
species_count |> slice(1:3) # keeps the first 3 rows
```

```
# A tibble: 3 x 10
```

	samplearea_id	`x-koord`	`y-koord`	lon	lat	year	route_length	median_elevation
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	268	2725500	1182500	9.08	46.8	1999	4444	1550
2	268	2725500	1182500	9.08	46.8	2000	4444	1550
3	268	2725500	1182500	9.08	46.8	2001	4444	1550

```
# i 2 more variables: n_visits <dbl>, n_recorded_species <dbl>
```

Choose rows using `slice_min()` or `slice_max()`

```
species_count |>
  slice_max(n_recorded_species, n = 3) # keeps the top 3 sites by n_recorded_species
```

A tibble: 5 x 10

	samplearea_id	`x-koord`	`y-koord`	lon	lat	year	route_length	median_elevation
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	422	2623500	1174500	7.75	46.7	2020	5065	1350
2	511	2707500	1270500	8.87	47.6	2007	6285	350
3	422	2623500	1174500	7.75	46.7	2019	5065	1350

i 2 more rows

i 2 more variables: n_visits <dbl>, n_recorded_species <dbl>

Choose rows using `slice_min()` or `slice_max()`

```
species_count |>
  slice_max(n_recorded_species, n = 3) # keeps the top 3 sites by n_recorded_species
```

A tibble: 5 x 10

	samplearea_id	`x-koord`	`y-koord`	lon	lat	year	route_length	median_elevation
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	422	2623500	1174500	7.75	46.7	2020	5065	1350
2	511	2707500	1270500	8.87	47.6	2007	6285	350
3	422	2623500	1174500	7.75	46.7	2019	5065	1350

i 2 more rows

i 2 more variables: n_visits <dbl>, n_recorded_species <dbl>

Pro tips:

- use `with_ties = FALSE`, to obtain the number of rows you want despite ties
- other slicing functions can be useful (e.g. `slice_head()`, `slice_sample()`)

Choose different rows using `distinct()`

```
species_count |> distinct(samplearea_id, lon, lat)
```

```
# A tibble: 267 x 3
  samplearea_id  lon  lat
      <dbl> <dbl> <dbl>
1          268  9.08  46.8
2          269  6.11  46.2
3          270  9.09  47.1
# i 264 more rows
```

Chaining multiple operations using the pipe |>

```
species_count |>  
  filter(year == 2025) |>  
  select(samplearea_id, year)
```

```
# A tibble: 261 x 2  
  samplearea_id year  
      <dbl> <dbl>  
1           268  2025  
2           269  2025  
3           270  2025  
# i 258 more rows
```

Chaining multiple operations using the pipe |>

```
species_count |>
  filter(year == 2025) |>
  select(samplearea_id, year)
```

```
# A tibble: 261 x 2
  samplearea_id year
      <dbl> <dbl>
1           268  2025
2           269  2025
3           270  2025
# i 258 more rows
```

NB:

- remember, this is interpreted by R as:
`select(filter(species_count, year == 2025), samplearea_id, year)`
- when chaining operations, keeping the data alone in the first line and indenting the rest improves the readability

Store your output into a reusable object using `<-` or `->`

Conventional way

```
species_count_2025 <- species_count |> filter(year == 2025)
```

Alternative way

```
species_count |> filter(year == 2025) -> species_count_2025
```

Pro tips:

- the conventional way helps to spot which objects are being created in the script
- when using many chained operation using the pipe `|>`, the alternative way lead to a more natural reading of the code

Sort rows using `arrange()`

```
species_count |> arrange(year)
```

```
# A tibble: 7,113 x 10
```

	samplearea_id	`x-koord`	`y-koord`	lon	lat	year	route_length	median_elevation
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	268	2725500	1182500	9.08	46.8	1999	4444	1550
2	269	2497500	1118500	6.11	46.2	1999	6232	450
3	270	2725500	1214500	9.09	47.1	1999	4081	1350

```
# i 7,110 more rows
```

```
# i 2 more variables: n_visits <dbl>, n_recorded_species <dbl>
```

Sort rows using `arrange()`

```
species_count |> arrange(year)
```

```
# A tibble: 7,113 x 10
```

	samplearea_id	`x-koord`	`y-koord`	lon	lat	year	route_length	median_elevation
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	268	2725500	1182500	9.08	46.8	1999	4444	1550
2	269	2497500	1118500	6.11	46.2	1999	6232	450
3	270	2725500	1214500	9.09	47.1	1999	4081	1350

```
# i 7,110 more rows
```

```
# i 2 more variables: n_visits <dbl>, n_recorded_species <dbl>
```

Pro tips:

- use multiple variables for hierarchical sorting
- use `desc()` for sorting by decreasing values (e.g. `desc(year)`)

Practice 5

Use `select()` and/or `filter()` and/or `slice()` and/or `arrange()` to determine:

1. the largest number of recorded species across all years
2. the largest number of recorded species in 2025

Create new columns using `mutate()`

```
species_count |>
  select(samplearea_id, route_length, year) |>
  filter(year == 2025) |>
  mutate(short_transect = route_length < mean(route_length))
```

A tibble: 261 x 4

	samplearea_id	route_length	year	short_transect
	<dbl>	<dbl>	<dbl>	<lgl>
1	268	4444	2025	TRUE
2	269	6232	2025	FALSE
3	270	4081	2025	TRUE

i 258 more rows

Create new columns using `mutate()`

```
species_count |>
  select(samplearea_id, route_length, year) |>
  filter(year == 2025) |>
  mutate(short_transect = route_length < mean(route_length))
```

A tibble: 261 x 4

	samplearea_id	route_length	year	short_transect
	<dbl>	<dbl>	<dbl>	<lgl>
1	268	4444	2025	TRUE
2	269	6232	2025	FALSE
3	270	4081	2025	TRUE

i 258 more rows

Pro tips:

- use `.before =` or `.after =` to control where the new column is added

Collapse groups into summary statistics using `summarise()`

```
species_count |>
  summarise(min_route_length = min(route_length),
            max_route_length = max(route_length))
```

```
# A tibble: 1 x 2
  min_route_length max_route_length
      <dbl>         <dbl>
1         1181         9411
```

Operations within groups

```
species_count |>  
  summarise(max_sp = max(n_recorded_species), .by = year)
```

```
# A tibble: 27 x 2
```

```
  year max_sp
```

```
<dbl> <dbl>
```

```
1  1999     53
```

```
2  2000     58
```

```
3  2001     59
```

```
# i 24 more rows
```

Operations within groups

```
species_count |>
  summarise(max_sp = max(n_recorded_species), .by = year)
```

```
# A tibble: 27 x 2
  year max_sp
<dbl> <dbl>
1  1999     53
2  2000     58
3  2001     59
# i 24 more rows
```

Pro tips:

- `.by =` works with the other verbs too, but is sometimes called `by` (e.g. in `slice()`)
- `.by =` is safer than `group_by()` which externalises the grouping

Collapse groups into list-columns using `summarise()` + `list()`

```
territories |>
  filter(territories_total > 0) |>
  summarise(sp = list(nameeg), .by = c(samplearea_id, year))
```

```
# A tibble: 7,113 x 3
  samplearea_id year sp
      <dbl> <dbl> <list>
1         279  2001 <chr [44]>
2         279  2002 <chr [46]>
3         279  2006 <chr [47]>
# i 7,110 more rows
```

Collapse groups into list-columns using `summarise()` + `list()`

```
territories |>
  filter(territories_total > 0) |>
  summarise(sp = list(nameeg), .by = c(samplearea_id, year))
```

```
# A tibble: 7,113 x 3
  samplearea_id year sp
      <dbl> <dbl> <list>
1         279  2001 <chr [44]>
2         279  2002 <chr [46]>
3         279  2006 <chr [47]>
# i 7,110 more rows
```

NB:

- the column `sp` is called a list-column

Line-by-line operations using `rowwise()`

```
territories |>
  filter(territories_total > 0) |>
  summarise(sp = list(nameeg), .by = c(samplearea_id, year)) |>
  rowwise() |>
  mutate(mute_swan = "Mute Swan" %in% sp) |>
  ungroup()
```

```
# A tibble: 7,113 x 4
```

	samplearea_id	year	sp	mute_swan
	<dbl>	<dbl>	<list>	<lgl>
1	279	2001	<chr [44]>	FALSE
2	279	2002	<chr [46]>	FALSE
3	279	2006	<chr [47]>	FALSE

```
# i 7,110 more rows
```

Line-by-line operations using `rowwise()`

```
territories |>
  filter(territories_total > 0) |>
  summarise(sp = list(nameeg), .by = c(samplearea_id, year)) |>
  rowwise() |>
  mutate(mute_swan = "Mute Swan" %in% sp) |>
  ungroup()
```

A tibble: 7,113 x 4

	samplearea_id	year	sp	mute_swan
	<dbl>	<dbl>	<list>	<lgl>
1	279	2001	<chr [44]>	FALSE
2	279	2002	<chr [46]>	FALSE
3	279	2006	<chr [47]>	FALSE

i 7,110 more rows

Pro tips:

- do not forget to call `ungroup()`, otherwise future calls will also be executed line by line, which can impact the results and be slow

Counting with `count()`

```
species_count |> count(year)
```

```
# A tibble: 27 x 2
```

```
  year      n
```

```
<dbl> <int>
```

```
1  1999    241
```

```
2  2000    264
```

```
3  2001    259
```

```
# i 24 more rows
```

Counting with `count()`

```
species_count |> count(year)
```

```
# A tibble: 27 x 2
```

```
  year      n
```

```
<dbl> <int>
```

```
1  1999    241
```

```
2  2000    264
```

```
3  2001    259
```

```
# i 24 more rows
```

Pro tips:

- you can use multiple variables here to obtain the number of rows for each distinct combination of values across those variables (as in `distinct()`)
- you can count using `summarise()` but `count()` is more direct
- you can sort the outcome (without `arrange()`) using the option `sort = TRUE`

Combining tables using `left_join()`

```
territories |> left_join(species_count)
```

Joining with `by = join\_by(samplearea\_id, year)`

```
# A tibble: 1,152,306 x 18
```

	species_id	eurung_id	nameeg	namelt	samplearea_id	year	territories_visit_1
	<dbl>	<dbl>	<chr>	<chr>	<dbl>	<dbl>	<dbl>
1	50	70	Little Grebe	Tachybaptus r~	268	1999	0
2	50	70	Little Grebe	Tachybaptus r~	268	2000	0
3	50	70	Little Grebe	Tachybaptus r~	268	2001	0

```
# i 1,152,303 more rows
```

```
# i 11 more variables: territories_visit_2 <dbl>, territories_visit_3 <dbl>,
```

```
# territories_total <dbl>, `x-koord` <dbl>, `y-koord` <dbl>, lon <dbl>, lat <dbl>,
```

```
# route_length <dbl>, median_elevation <dbl>, n_visits <dbl>, n_recorded_species <dbl>
```

Combining tables using `left_join()`

```
territories |> left_join(species_count)
```

Joining with `by = join\_by(samplearea\_id, year)`

```
# A tibble: 1,152,306 x 18
```

	species_id	euring_id	nameeg	namelt	samplearea_id	year	territories_visit_1
	<dbl>	<dbl>	<chr>	<chr>	<dbl>	<dbl>	<dbl>
1	50	70	Little Grebe	Tachybaptus r~	268	1999	0
2	50	70	Little Grebe	Tachybaptus r~	268	2000	0
3	50	70	Little Grebe	Tachybaptus r~	268	2001	0

```
# i 1,152,303 more rows
```

```
# i 11 more variables: territories_visit_2 <dbl>, territories_visit_3 <dbl>,  
# territories_total <dbl>, `x-koord` <dbl>, `y-koord` <dbl>, lon <dbl>, lat <dbl>,  
# route_length <dbl>, median_elevation <dbl>, n_visits <dbl>, n_recorded_species <dbl>
```

NB:

- for `left_join()` to work automatically, some columns with identical names must exist in both datasets (called keys)

Practice 6

Use the functions we have seen to obtain:

1. the range of median elevation according to the number of visits
2. the 3 most recorded species for each year
3. the number of observations for *Sturnus vulgaris* for each year

Reshaping data from long to wide format using `pivot_wider()`

Let's create an "site x species" detection table for 2025:

```
territories |>
  filter(year == 2025) |>
  select(samplearea_id, nameeg, territories_total) |>
  pivot_wider(names_from = nameeg,
              values_from = territories_total)
```

```
# A tibble: 261 x 163
```

	samplearea_id	`Little Grebe`	`Great Crested Grebe`	`Grey Heron`	`Little Bittern`
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	268	0	0	0	0
2	269	0	0	0	0
3	270	0	0	0	0

```
# i 258 more rows
```

```
# i 158 more variables: `White Stork` <dbl>, `Mute Swan` <dbl>, `Greylag Goose` <dbl>,
# `Egyptian Goose` <dbl>, `Mallard` <dbl>, `Red-crested Pochard` <dbl>,
# `Tufted Duck` <dbl>, `Common Merganser` <dbl>, `European Honey Buzzard` <dbl>,
# `Red Kite` <dbl>, `Black Kite` <dbl>, `Eurasian Goshawk` <dbl>,
# `Eurasian Sparrowhawk` <dbl>, `Common Buzzard` <dbl>, `Eurasian Hobby` <dbl>,
# `Common Kestrel` <dbl>, `Black Grouse` <dbl>, `Rock Ptarmigan` <dbl>, ...
```

Practice 7

Try to use `pivot_longer()` to get back to the original long format of the data from the output of the following code:

```
territories |>
  filter(year == 2025) |>
  select(samplearea_id, nameeg, territories_total) |>
  pivot_wider(names_from = nameeg,
              values_from = territories_total)
```

Wrangling spatial data

The simple feature package {sf}

{sf} is a package developed to handle spatial data which plays along with the {tidyverse} (it is inspired by PostGIS, see e.g. postgis.net/workshops/postgis-intro/index.html)

```
library(sf)
```

Linking to GEOS 3.14.1, GDAL 3.12.4, PROJ 9.8.1; sf_use_s2() is TRUE

```
species_count |>  
  st_as_sf(coords = c("lon", "lat"), crs = 4326) -> species_count_sf
```

NB:

- the order of the columns provided to `coords =` matters (see next)
- the coordinate reference system (`crs`) is here provided as a so-called EPSG code (4326 = WGS 84)

Simple feature collections

```
species_count_sf
```

Simple feature collection with 7113 features and 8 fields

Geometry type: POINT

Dimension: XY

Bounding box: xmin: 6.110458 ymin: 45.8476 xmax: 10.40901 ymax: 47.72377

Geodetic CRS: WGS 84

A tibble: 7,113 x 9

	samplearea_id	`x-koord`	`y-koord`	year	route_length	median_elevation	n_visits
*	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	268	2725500	1182500	1999	4444	1550	3
2	268	2725500	1182500	2000	4444	1550	3
3	268	2725500	1182500	2001	4444	1550	3

i 7,110 more rows

i 2 more variables: n_recorded_species <dbl>, geometry <POINT [°]>

Pro tips:

- check carefully the bounding box to make sure that latitudes and longitudes are OK
- simple feature collections are tibbles with a specific list-column called `geometry`

Computing the total surface area

```
species_count_sf |>
  summarise(geometry = st_combine(geometry)) |> # all points are brought together
  mutate(hull = st_convex_hull(geometry), # create shapes encompassing all points
         area_m2 = st_area(hull), # compute the area in m^2
         area_km2 = units::set_units(area_m2, "km^2")) |> # express area in km^2
  select(-hull) |> # drop the hull
  st_drop_geometry() # drop the geometry
```

```
# A tibble: 1 x 2
  area_m2 area_km2
*      [m^2]   [km^2]
1 48995663682.  48996.
```

Computing the total surface area

```
species_count_sf |>
  summarise(geometry = st_combine(geometry)) |> # all points are brought together
  mutate(hull = st_convex_hull(geometry), # create shapes encompassing all points
         area_m2 = st_area(hull), # compute the area in m^2
         area_km2 = units::set_units(area_m2, "km^2")) |> # express area in km^2
  select(-hull) |> # drop the hull
  st_drop_geometry() # drop the geometry
```

```
# A tibble: 1 x 2
  area_m2 area_km2
*      [m^2]   [km^2]
1 48995663682.  48996.
```

- the syntax `namespace::function` allows one to use a function from a package without calling `library()`
- you must use `st_drop_geometry()` to drop the geometry column from a sf collection; `select()` does not remove the protected geometry column

Computing the distance of each site to the center of the studied area

```
species_count_sf |>
  distinct(samplearea_id, geometry) |>
  mutate(centroid = st_centroid(st_combine(geometry)),
         dist = st_distance(centroid, geometry, by_element = TRUE))
```

Simple feature collection with 267 features and 2 fields

Active geometry column: geometry

Geometry type: POINT

Dimension: XY

Bounding box: xmin: 6.110458 ymin: 45.8476 xmax: 10.40901 ymax: 47.72377

Geodetic CRS: WGS 84

A tibble: 267 x 4

	samplearea_id	geometry	centroid	dist
*	<dbl>	<POINT [°]>	<POINT [°]>	[m]
1	268	(9.082159 46.78185)	(8.234991 46.79076)	64509.
2	269	(6.110458 46.21013)	(8.234991 46.79076)	174955.
3	270	(9.090984 47.06963)	(8.234991 46.79076)	72016.

i 264 more rows

NB: if a computation seems to take a very long time, it is *often* a sign that you are not handling the problem properly

Practice 8

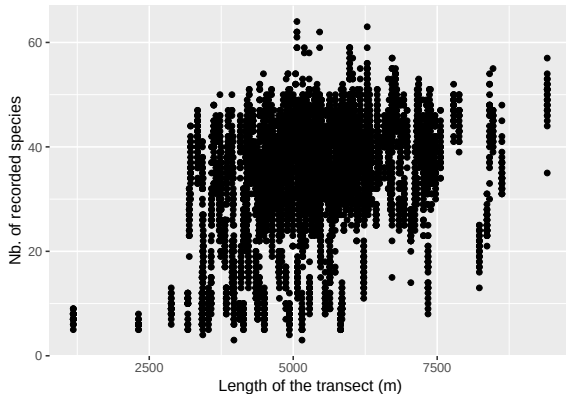
Use the functions we have seen to obtain:

1. the sampled area (sensu convex hull) across years
2. the IDs of the sampled areas which are most peripheral

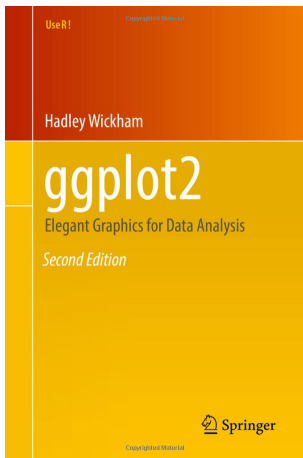
Plotting data

Plots are built in `{ggplot2}` by adding components together

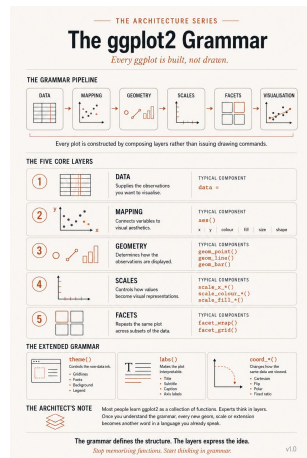
```
ggplot(species_count) +  
  geom_point(aes(y = n_recorded_species, x = route_length)) +  
  labs(y = "Nb. of recorded species", x = "Length of the transect (m)")
```



The grammar of graphics plotting framework



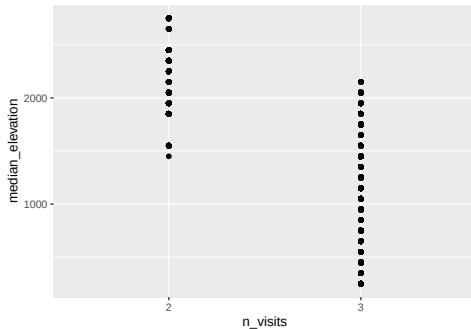
source: <https://ggplot2-book.org>



source: <https://linkedin.com/in/adalbert-ngongang-phd-637995a3>

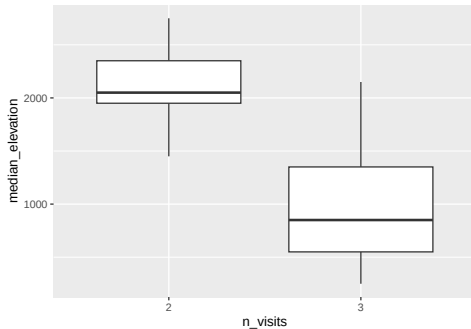
Geoms are the fundamental building blocks

```
species_count |>  
  mutate(n_visits = factor(n_visits)) |>  
  ggplot() +  
    geom_point(aes(y = median_elevation, x = n_visits))
```



Geoms are the fundamental building blocks

```
species_count |>  
  mutate(n_visits = factor(n_visits)) |>  
  ggplot() +  
    geom_boxplot(aes(y = median_elevation, x = n_visits))
```

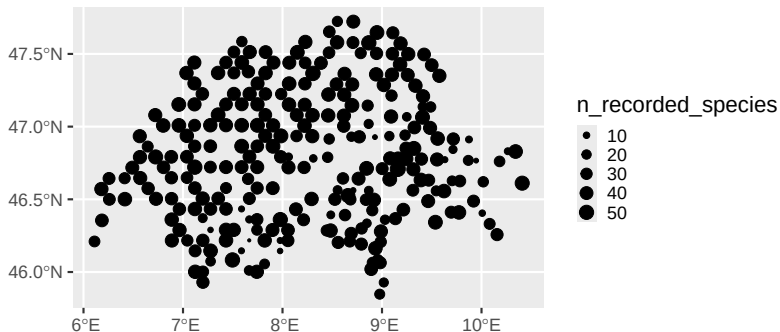


Practice 9

1. check the possible geoms in the official ggplot2 cheat sheet (accessible via Help)
2. explore briefly r-graph-gallery.com

Objects created with {sf} can be plotted using {ggplot2} too

```
species_count_sf |>  
  filter(year == 2025) |>  
  ggplot() +  
    geom_sf(aes(size = n_recorded_species)) # geometry = geometry is assumed
```



Understanding aesthetic mappings

- `aes()` is used to defined how data connect to plotting elements via aesthetics
- different geom expect different mandatory and optional aesthetics
(see e.g. `?geom_point`)

Understanding aesthetic mappings

- `aes()` is used to defined how data connect to plotting elements via aesthetics
- different geom expect different mandatory and optional aesthetics (see e.g. `?geom_point`)
- `aes()` is not the place to force colours, sizes, etc.; it only forces relationship:

Bad

```
ggplot(species_count) +  
  geom_point(aes(y = n_recorded_species, x = route_length, colour = "blue"))
```

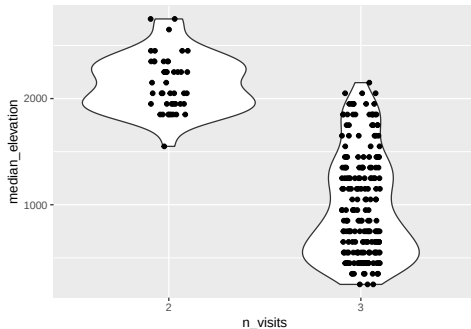
Good

```
ggplot(species_count) +  
  geom_point(aes(y = n_recorded_species, x = route_length), colour = "blue")
```

Understanding aesthetic mappings

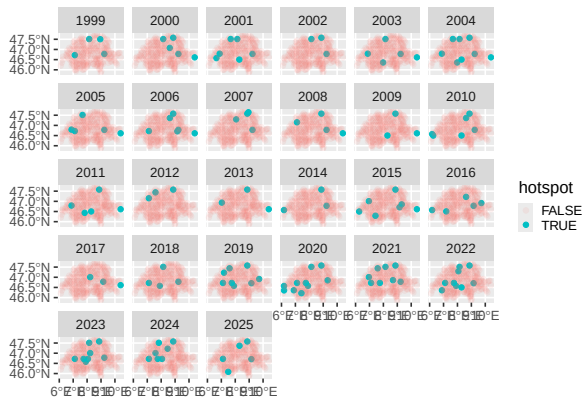
- `aes()` can be located outside a geom to apply to several of them at once:

```
species_count |>  
  filter(year == 2025) |>  
  mutate(n_visits = factor(n_visits)) |>  
  ggplot(aes(y = median_elevation, x = n_visits)) +  
    geom_violin() +  
    geom_jitter(height = 0, width = 0.1)
```



Plot different subsets using faceting

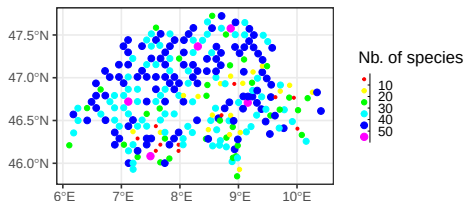
```
species_count_sf |>
  mutate(hotspot = factor(n_recorded_species > 50)) |>
  ggplot() +
    geom_sf(aes(colour = hotspot, alpha = hotspot)) +
    facet_wrap(~ year)
```



Making plots beautiful & readable

Much can be achieved by using a better theme and defining scales:

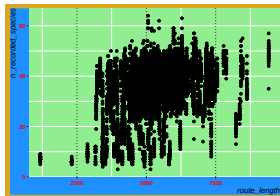
```
species_count_sf |>
  filter(year == 2025) |>
  ggplot() +
    geom_sf(aes(size = n_recorded_species,
                 colour = n_recorded_species)) +
    scale_color_binned(palette = rainbow, name = "Nb. of species") +
    scale_size_binned(range = c(0.5, 5), name = "Nb. of species") +
    guides(size = guide_bins(), color = guide_bins()) + # unusually needed here
    theme_bw(base_size = 25)
```



Making plots beautiful & readable

To modify the theme yourself, use `theme_sub_*()` or (`theme()`) in combination with the appropriate `element_*()` function:

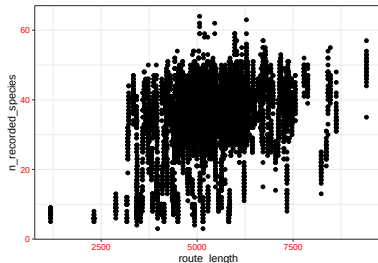
```
ggplot(species_count) +  
  geom_point(aes(y = n_recorded_species, x = route_length)) +  
  theme_sub_axis(text = element_text(size = 15, face = "bold", colour = "red"),  
                 title = element_text(face = "italic", hjust = 1)) +  
  theme_sub_panel(grid.major.x = element_line(linetype = "dotted", colour = "black"),  
                 background = element_rect(fill = "lightgreen")) +  
  theme_sub_plot(background = element_rect(fill = "dodgerblue", colour = "goldenrod",  
                                           linewidth = 10))
```



Making plots beautiful & readable

You can define a common theme to be used for all your plots:

```
(theme_bw(base_size = 20) +  
  theme_sub_axis(text = element_text(colour = "red")) |>  
  theme_set())  
  
ggplot(species_count) +  
  geom_point(aes(y = n_recorded_species, x = route_length))
```



NB: the extra `()` are needed since `|>` binds more tightly than `+`

Practice 10

- combine data wrangling and plotting to illustrate the change in species abundance across years for the top 6 most encountered species

A dash of base R

What is base R?

This is not base R (it relies on {tidyverse}, an external R package)

```
species_count |>  
  summarise(mean_nb_sp = mean(n_recorded_species), .by = year)
```

```
# A tibble: 27 x 2
```

```
  year mean_nb_sp
```

```
<dbl>      <dbl>
```

```
1  1999      31.4
```

```
2  2000      32.5
```

```
3  2001      32.7
```

```
# i 24 more rows
```

What is base R?

This is typical base R (it does not rely on an external R package)

```
results <- data.frame(year = unique(species_count$year), mean_nb_sp = NA)
for (y in unique(species_count$year)) {
  n_recorded_species_year <- species_count$n_recorded_species[species_count$year == y]
  results[results$year == y, "mean_nb_sp"] <- mean(n_recorded_species_year)
}
head(results, n = 3)
```

	year	mean_nb_sp
1	1999	31.36515
2	2000	32.48485
3	2001	32.70270

NB:

- it looks more complicated (but from a developer's perspective, it is actually clearer)

What is base R?

But this is also base R (and a little closer to {tidyverse} in spirit)

```
results <- with(species_count,  
                aggregate(list(mean_nb_sp = n_recorded_species),  
                          by = list(year = year), FUN = mean))  
head(results, n = 3)
```

	year	mean_nb_sp
1	1999	31.36515
2	2000	32.48485
3	2001	32.70270

NB:

- there are often plenty of alternatives to achieve the same output

Some key objects in base R

object	typical number of dimensions	content restriction	usage example
the <i>vector</i>	1	all elements must have same type	storing data for a variable
the <i>list</i>	tree-like structure	each list can be anything	output of model fit
the <i>matrix</i>	2	elements must have same type everywhere	design matrix of a linear model
the <i>array</i>	N	elements must have same type everywhere	posteriors in Bayesian analyses
the <i>data frame</i> or <i>tibble</i>	2	elements must have same type in a column	input data for a model

Some key objects in base R

object	typical number of dimensions	content restriction	usage example
the <i>vector</i>	1	all elements must have same type	storing data for a variable
the <i>list</i>	tree-like structure	each list can be anything	output of model fit
the <i>matrix</i>	2	elements must have same type everywhere	design matrix of a linear model
the <i>array</i>	N	elements must have same type everywhere	posteriors in Bayesian analyses
the <i>data frame</i> or <i>tibble</i>	2	elements must have same type in a column	input data for a model

NB:

- the different objects are generally created by functions, so you won't have to worry about creating them
- what you need to worry about is how to extract information from each of those objects

Extracting elements from vectors

```
some_vector <- letters # we create a small vector  
some_vector
```

```
[1] "a" "b" "c" "d" "e" "f" "g" "h" "i" "j" "k" "l" "m" "n" "o" "p" "q" "r" "s" "t" "u"  
[22] "v" "w" "x" "y" "z"
```

```
some_vector[c(1, 12, 5, 24)]
```

```
[1] "a" "l" "e" "x"
```

```
some_vector[some_vector %in% c("a", "l", "e", "x")]
```

```
[1] "a" "e" "l" "x"
```

```
some_namedvector <- c(a = 1, b = 5)  
some_namedvector["b"]
```

```
b  
5
```

Extracting elements from lists

```
some_list <- list(1, 1:5) # we create a small list  
some_list
```

```
[[1]]  
[1] 1
```

```
[[2]]  
[1] 1 2 3 4 5
```

```
some_list[2]
```

```
[[1]]  
[1] 1 2 3 4 5
```

```
some_list[[2]]
```

```
[1] 1 2 3 4 5
```

```
some_list[[2]][2]
```

```
[1] 2
```

Extracting elements from named lists

```
some_namedlist <- list(a = 1, b = 1:5) # we create a small named list  
some_namedlist["b"]
```

```
$b  
[1] 1 2 3 4 5
```

```
some_namedlist[["b"]]
```

```
[1] 1 2 3 4 5
```

```
some_namedlist$b
```

```
[1] 1 2 3 4 5
```

Extracting elements from matrices

```
some_matrix <- matrix(1:9, nrow = 3) # we create a small matrix
some_matrix
```

```
      [,1] [,2] [,3]
[1,]     1     4     7
[2,]     2     5     8
[3,]     3     6     9
```

```
some_matrix[2, ]
```

```
[1] 2 5 8
```

```
some_matrix[, 2]
```

```
[1] 4 5 6
```

```
some_matrix[2, 2]
```

```
[1] 5
```

Extracting elements from matrices

```
some_matrix[c(1, 3), c(1, 3)]
```

```
      [,1] [,2]  
[1,]     1     7  
[2,]     3     9
```

```
some_matrix[cbind(c(1, 3), c(1, 3))]
```

```
[1] 1 9
```


Extracting elements from named matrices

```
colnames(some_matrix) <- paste0("col_", 1:3) # we name the columns
rownames(some_matrix) <- paste0("row_", 1:3) # we name the rows
some_matrix
```

	col_1	col_2	col_3
row_1	1	4	7
row_2	2	5	8
row_3	3	6	9

```
some_matrix["row_2", "col_3"]
```

```
[1] 8
```

Extracting elements from arrays

```
some_array <- array(1:30, dim = c(5, 3, 2)) # we create a small 3D array
```

```
some_array
```

```
, , 1
     [,1] [,2] [,3]
[1,]    1    6   11
[2,]    2    7   12
[3,]    3    8   13
[4,]    4    9   14
[5,]    5   10   15
```

```
, , 2
     [,1] [,2] [,3]
[1,]   16   21   26
[2,]   17   22   27
[3,]   18   23   28
[4,]   19   24   29
[5,]   20   25   30
```

```
some_array[2, , ] # rows
```

```
     [,1] [,2]
[1,]     2   17
[2,]     7   22
[3,]    12   27
```

```
some_array[, 2, ] # columns
```

```
     [,1] [,2]
[1,]     6   21
[2,]     7   22
[3,]     8   23
[4,]     9   24
[5,]    10   25
```

```
some_array[, , 2] # depth
```

```
     [,1] [,2] [,3]
[1,]   16   21   26
[2,]   17   22   27
[3,]   18   23   28
[4,]   19   24   29
[5,]   20   25   30
```

```
some_array[2, 2, 2] # cell
```

```
[1] 22
```

```
some_array[2, , 2] # row 2
                  # at depth 2
```

```
[1] 17 22 27
```

Extracting elements from data frames / tibbles

```
species_count[1, ]
```

```
# A tibble: 1 x 10
```

	samplearea_id	`x-koord`	`y-koord`	lon	lat	year	route_length	median_elevation
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	268	2725500	1182500	9.08	46.8	1999	4444	1550

```
# i 2 more variables: n_visits <dbl>, n_recorded_species <dbl>
```

```
species_count[, 1]
```

```
# A tibble: 7,113 x 1
```

	samplearea_id
	<dbl>

1	268
---	-----

2	268
---	-----

3	268
---	-----

```
# i 7,110 more rows
```

Extracting elements from data frames / tibbles

```
species_count[, "year"]
```

```
# A tibble: 7,113 x 1
```

```
  year
```

```
<dbl>
```

```
1  1999
```

```
2  2000
```

```
3  2001
```

```
# i 7,110 more rows
```

```
species_count["year"]
```

```
# A tibble: 7,113 x 1
```

```
  year
```

```
<dbl>
```

```
1  1999
```

```
2  2000
```

```
3  2001
```

```
# i 7,110 more rows
```

Extracting elements from data frames / tibbles

```
species_count_small <- slice(species_count, 1:10) # for smaller data  
species_count_small[["year"]]
```

```
[1] 1999 2000 2001 2002 2003 2004 2005 2006 2007 2008
```

```
species_count_small$year
```

```
[1] 1999 2000 2001 2002 2003 2004 2005 2006 2007 2008
```

A few very useful base R functions I would be sad to work without

Check the following on your own time:

- `help()` for help (or `?` for short)
- `plot()` for fast & dirty plots
- `str()` for inspecting objects
- `dput()` for sharing a reproducible example when asking for help (useful for LLMs)
- `function()` for creating a function (or `\()` for short)
- `debugonce()` for debugging a function line by line
- `rnorm()`, `runif()` for simulating data
- `set.seed()` for “fixing” randomness
- `replicate()` for doing something several times

Pro tips:

- check them out in your own time; they are worth learning about

House keeping

When and what to update

- update R packages regularly using the RStudio “Packages tab”
(at least once a month)
- update R a few times a year:
check www.r-project.org (May and October tend to be good times)
- update RStudio at least after each update of R

Learning more

Getting help

- don't forget that functions have help pages associated to them
- R for Data Science free book: r4ds.hadley.nz
- there is an active Slack channel: dslc.io
- talk to each other
- LLMs?
if you do, don't blindly copy/paste, and beware:
LLMs make mistakes and make things up

Getting better

- learn “base R” too (e.g. cran.r-project.org/doc/manuals/r-release/R-intro.html)
- try to understand why things do not behave as you expect by experimenting & avoid always using a workaround
- do use LLMs to review your code (rather than to create it)
the more you know, the more LLMs will be helpful to you!

Questions?